

MFG 510 Assignment 2: Forecasting (TES) and Linear Programming

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Problem 1: Triple exponential smoothing

Sales: $A_1 = 40$, $A_2 = 50$, $A_3 = 60$, $A_4 = 50$, $A_5 = 44$. $\alpha = \beta = \gamma = 0.2$, $N = 4$.

$$F_t = \alpha A_t / c_{t-N} + (1 - \alpha)(F_{t-1} + T_{t-1}), \quad T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}, \quad c_t = \gamma A_t / F_t + (1 - \gamma)c_{t-N}, \quad f(t+\tau) = (F_t + \tau T_t) \cdot c_{t+\tau-N}.$$

(a) Initialization on the first year

$$F_4 = (40 + 50 + 60 + 50)/4 = 200/4 = \mathbf{50.0000}$$

$$T_4 = \mathbf{0}$$

i	A_i	$c_i = A_i/F_4$
1	40	$40/50 = \mathbf{0.8000}$
2	50	$50/50 = \mathbf{1.0000}$
3	60	$60/50 = \mathbf{1.2000}$
4	50	$50/50 = \mathbf{1.0000}$

Check: the four factors add up to $N = 4$ ($0.8 + 1.0 + 1.2 + 1.0 = 4.0$).

(b) First forecast and its error

The forecast for $t = 5$ is made at the end of $t = 4$ with $\tau = 1$, so it uses $c_{5-4} = c_1$ (the Q1 factor):

$$f_5 = (F_4 + 1 \cdot T_4) \cdot c_1 = (50 + 0)(0.8000) = \mathbf{40.0000}$$

$$\varepsilon_5 = f_5 - A_5 = 40 - 44 = \mathbf{-4.0000} \text{ (the forecast was 4 thousand units too low).}$$

(c) Update with $A_5 = 44$

$$F_5 = \alpha A_5 / c_1 + (1 - \alpha)(F_4 + T_4) = 0.2(44/0.8000) + 0.8(50 + 0) = 0.2(55) + 40 = 11 + 40 = \mathbf{51.0000}$$

$$T_5 = \beta(F_5 - F_4) + (1 - \beta)T_4 = 0.2(51 - 50) + 0.8(0) = \mathbf{0.2000}$$

$$c_5 = \gamma A_5 / F_5 + (1 - \gamma)c_1 = 0.2(44/51) + 0.8(0.8000) = 0.2(0.8627) + 0.6400 = 0.1725 + 0.6400 = \mathbf{0.8125}$$

c_5 is the updated Q1 factor. It replaces c_1 and will be used for the next Q1 forecast ($t = 9$).

(d) Forecasts made at the end of $t = 5$

$$f_6 = (F_5 + 1 \cdot T_5) \cdot c_{6-4} = (51 + 0.2) \cdot c_2 = 51.2(1.0000) = \mathbf{51.2000}$$

$$f_7 = (F_5 + 2 \cdot T_5) \cdot c_{7-4} = (51 + 0.4) \cdot c_3 = 51.4(1.2000) = \mathbf{61.6800}$$

Which factors and why. $t = 6$ is Q2 of year 2, and $t = 7$ is Q3 of year 2. The seasonal factor must match the quarter being forecast, so each forecast uses the most recent factor for that same quarter, which is the one from exactly one season ($N = 4$ periods) earlier: c_2 for Q2 and c_3 for Q3. The new factor c_5 belongs to Q1 and cannot be used for Q2 or Q3.

Problem 2: Product mix by the graphical method

(a) Formulation

X_1 = units of Product A made per week, X_2 = units of Product B made per week.

Maximize $Z = 1.5X_1 + 1.5X_2$

subject to

Constraint	Expression
Molding	$3X_1 + 2X_2 \leq 600$
Painting	$2X_1 + 4X_2 \leq 720$
Packing	$2X_1 + 3X_2 \leq 480$
Nonnegativity	$X_1, X_2 \geq 0$

(b) Feasible region

Axis intercepts of each constraint line (set the other variable to 0):

Line	X_1 intercept	X_2 intercept
Molding $3X_1 + 2X_2 = 600$	$600/3 = 200$	$600/2 = 300$
Painting $2X_1 + 4X_2 = 720$	$720/2 = 360$	$720/4 = 180$
Packing $2X_1 + 3X_2 = 480$	$480/2 = 240$	$480/3 = 160$

On the X_1 axis the tightest limit is molding (200). On the X_2 axis it is packing (160). The remaining corner is where molding and packing cross:

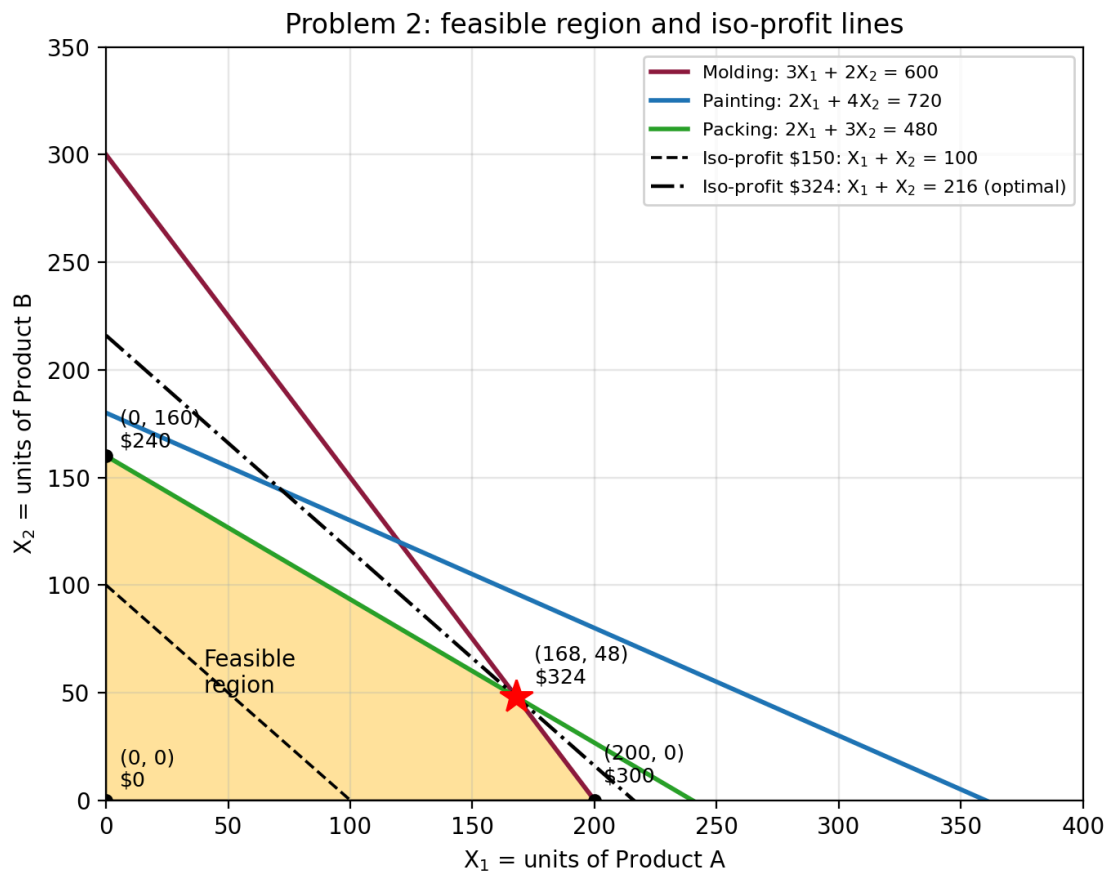
$$3X_1 + 2X_2 = 600 \quad (\times 3) \rightarrow 9X_1 + 6X_2 = 1800$$

$$2X_1 + 3X_2 = 480 \quad (\times 2) \rightarrow 4X_1 + 6X_2 = 960$$

$$\text{Subtract: } 5X_1 = 840 \rightarrow X_1 = 168, X_2 = (600 - 3 \cdot 168)/2 = (600 - 504)/2 = 48.$$

Check painting at (168, 48): $2(168) + 4(48) = 528 \leq 720$, so the point is feasible.

Corner points: **(0, 0), (200, 0), (168, 48), (0, 160).**



(c) Iso-profit line and optimum

Chosen profit \$150: $1.5X_1 + 1.5X_2 = 150$, i.e. $X_1 + X_2 = 100$ (dashed line, slope -1). Moving this line parallel to itself away from the origin, it last touches the feasible region at $(168, 48)$. That happens because the iso-profit slope (-1) lies between the molding slope ($-3/2$) and the packing slope ($-2/3$). At that point the line is $X_1 + X_2 = 216$, so $Z = 1.5(216) = \$324$.

Corner check: $Z(0, 0) = \$0$, $Z(200, 0) = \$300$, **$Z(168, 48) = \324** , $Z(0, 160) = \$240$.

Optimal mix: 168 units of A and 48 units of B per week, for a maximum weekly contribution of \$324.

Resource	Used at optimum	Available	Slack	Binding?
Molding	$3(168) + 2(48) = 600$	600	0	Yes
Painting	$2(168) + 4(48) = 528$	720	192	No
Packing	$2(168) + 3(48) = 480$	480	0	Yes

Painting is never binding. Its line (intercepts 360 and 180) lies completely outside the feasible region. The most painting any feasible mix can use is $2(0) + 4(160) = 640$ minutes at $(0, 160)$, so there are always at least 80 idle painting minutes, and 192 idle minutes at the optimum. For the manager, this means extra painting capacity is worth \$0: adding painting hours, overtime or equipment would not raise profit, and the 192 spare minutes could go to other work. Molding and packing are the bottlenecks, and any capacity investment should go there.

Problem 3: Rienzi Farms by the corner-point method

(a) Formulation

X_1 = acres of soybeans, X_2 = acres of sugar cane.

Maximize $Z = 1,000X_1 + 2,500X_2$

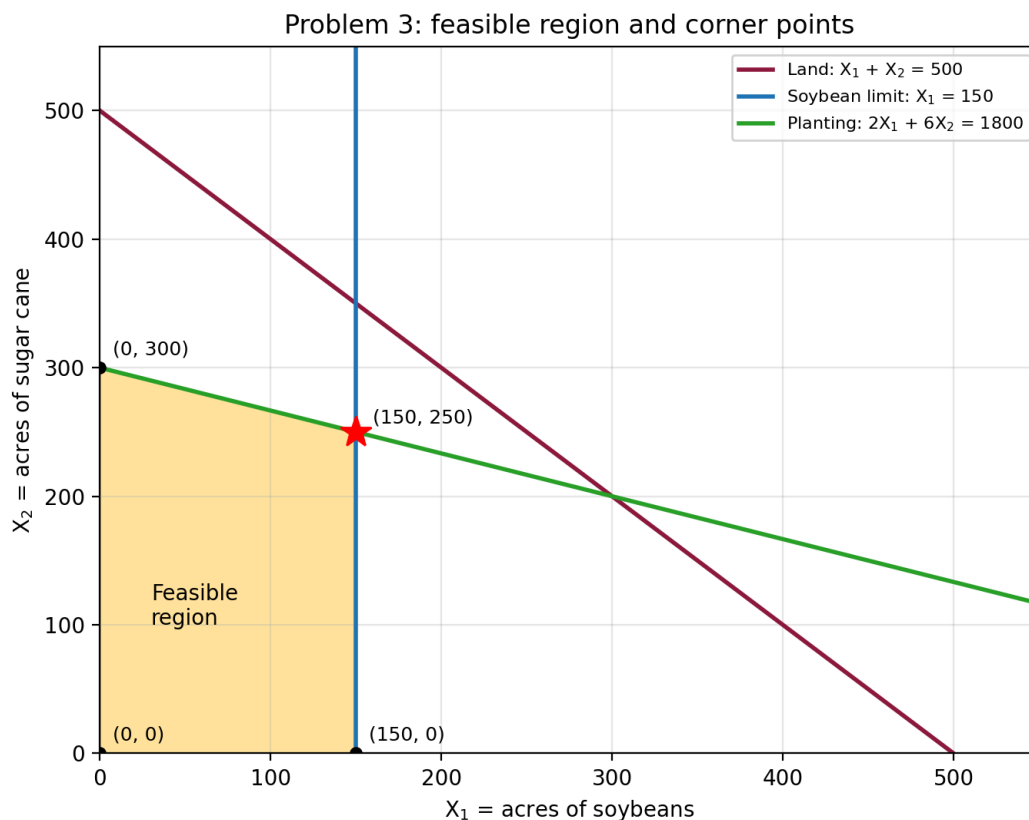
subject to

Constraint	Expression
Land	$X_1 + X_2 \leq 500$
Soybean program limit	$X_1 \leq 150$
Planting time	$2X_1 + 6X_2 \leq 1,800$
Nonnegativity	$X_1, X_2 \geq 0$

(b) Corner-point method

Planting-time line intercepts: $X_1 = 1800/2 = 900$, $X_2 = 1800/6 = 300$. Land line intercepts: 500 and 500.

The land line and the planting line would cross at $X_1 + X_2 = 500$ and $2X_1 + 6X_2 = 1800$. Subtracting $2 \times (\text{land})$ from planting gives $4X_2 = 800$, so $X_2 = 200$ and $X_1 = 300$. That violates $X_1 \leq 150$, so this crossing is not a corner of the feasible region, and **land never binds**.



Corner	Defining constraints	Calculation	Z
(0, 0)	$X_1 = 0, X_2 = 0$	$0 + 0$	\$0
(150, 0)	$X_1 = 150, X_2 = 0$	$1,000(150) + 0$	\$150,000
(150, 250)	$X_1 = 150, 2X_1 + 6X_2 = 1,800$ $\rightarrow 6X_2 = 1,500$	$1,000(150) + 2,500(250)$	\$775,000
(0, 300)	$X_1 = 0, 2X_1 + 6X_2 = 1,800 \rightarrow$ $X_2 = 300$	$0 + 2,500(300)$	\$750,000

Feasibility check at (150, 250): land $150 + 250 = 400 \leq 500$, planting $2(150) + 6(250) = 1,800 \leq 1,800$.

Optimal plan: 150 acres of soybeans and 250 acres of sugar cane, for a total profit of \$775,000.

Unplanted land: $500 - (150 + 250) = 100$ acres. Planting time (1,800 hours) and the soybean limit (150 acres) run out before the land does.

Problem 4: Rollins Publishing

(a) Formulation

Decision variables for books $i = 1, \dots, 5$:

- X_i = number of copies of book i printed (continuous, ≥ 0)
- $Y_i = 1$ if book i is published, 0 otherwise

Unit contribution $S_i - V_i$: book 1: $40 - 19 = 21$, book 2: $60 - 28 = 32$, book 3: $52 - 30 = 22$, book 4: $34 - 20 = 14$, book 5: $45 - 20 = 25$.

Objective. Maximize $Z = \sum_i [(S_i - V_i)X_i - F_i Y_i]$

Maximize $Z = 21X_1 + 32X_2 + 22X_3 + 14X_4 + 25X_5 - 12,000Y_1 - 21,000Y_2 - 15,000Y_3 - 10,000Y_4 - 18,000Y_5$

subject to

Constraint	Expression
Capacity	$X_1 + X_2 + X_3 + X_4 + X_5 \leq 20,000$
Link / demand, book 1	$X_1 \leq 9,000Y_1$
Link / demand, book 2	$X_2 \leq 8,000Y_2$
Link / demand, book 3	$X_3 \leq 5,000Y_3$
Link / demand, book 4	$X_4 \leq 6,000Y_4$
Link / demand, book 5	$X_5 \leq 7,000Y_5$
Domains	$X_i \geq 0, Y_i \in \{0, 1\}$ for $i = 1, \dots, 5$

The linking constraint $X_i \leq D_i Y_i$ does two jobs. If $Y_i = 0$ it forces $X_i = 0$ (no copies without publishing). If $Y_i = 1$ it caps copies at the maximum demand D_i and charges the fixed cost F_i in the objective.

(b) Spot the error

With $X_i \geq D_i Y_i$, setting $Y_i = 0$ gives only $X_i \geq 0$, so the model can print any number of copies without paying the fixed cost. Since the objective rewards $Y_i = 0$ (it removes $-F_i$), the solver sets every $Y_i = 0$ and prints freely, for example 20,000 copies of book 2 for $32 \times 20,000 = \$640,000$ with no fixed cost at all. Nothing caps the copies at the maximum demand either, and choosing $Y_i = 1$ would force at least D_i copies instead of at most D_i . The link is reversed: it should be $X_i \leq D_i Y_i$.

Problem 5: Machine shop with downtime and defects

Decision variables

X_A = units of Product A **started** per week, X_B = units of Product B **started** per week.

(i) Downtime: available minutes per step

Step	Downtime	Available minutes
Polishing	5%	$2,400(1 - 0.05) = 2,280$
Drilling	10%	$2,400(1 - 0.10) = 2,160$
Tapping	20%	$2,400(1 - 0.20) = 1,920$

(ii) Defects: units reaching each step

Defective parts are scrapped after the step where they are found, so each step only processes the units that survived the earlier steps ($\alpha = 0.01$, $\beta = 0.05$, $\gamma = 0.05$):

Point in the line	Fraction of units started	Product A	Product B
Reach polishing	1	X_A	X_B
Reach drilling	$(1 - \alpha) = 0.99$	$0.99X_A$	$0.99X_B$
Reach tapping	$(1 - \alpha)(1 - \beta) = 0.99(0.95) = 0.9405$	$0.9405X_A$	$0.9405X_B$
Good units out	$(1 - \alpha)(1 - \beta)(1 - \gamma) = 0.9405(0.95) = 0.893475$	$0.893475X_A$	$0.893475X_B$

Capacity constraints (minutes per unit $\times$ units reaching the step $\leq$ available minutes)

Polishing: $5X_A + 2X_B \leq 2,280$

Drilling: $2(0.99)X_A + 5(0.99)X_B \leq 2,160 \rightarrow 1.98X_A + 4.95X_B \leq 2,160$

Tapping: $2(0.9405)X_A + 3(0.9405)X_B \leq 1,920 \rightarrow 1.881X_A + 2.8215X_B \leq 1,920$

(iii) Objective: profit on good units only

$$\text{Maximize } Z = 5(0.893475)X_A + 3(0.893475)X_B = \mathbf{4.467375X_A + 2.680425X_B}$$

Complete model

$$\text{Maximize } Z = 4.467375X_A + 2.680425X_B$$

subject to

Constraint	Expression
Polishing	$5X_A + 2X_B \leq 2,280$
Drilling	$1.98X_A + 4.95X_B \leq 2,160$
Tapping	$1.881X_A + 2.8215X_B \leq 1,920$
Nonnegativity	$X_A, X_B \geq 0$